

Jakub Sokal

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Profile Summary

Software engineer with 12 months of commercial experience diagnosing and resolving a live double-charge defect in a customer billing system and owning end-to-end Stripe and SEPA integrations at AMCS Group. Named student researcher on the EU-funded Prompt Me project at Lero, University of Limerick, designing and building a multi-agent LLM orchestration framework for AI-driven Requirements Engineering (RE). Graduating August 2026, open to graduate roles in backend engineering, full-stack development, and fintech.

Education

University of Limerick

Limerick, Co. Limerick

Bachelor of Science in Computer Science | Expected August 2026 | First Class Honours (QCA: 3.44/4.0)

Course Work: Database Systems, Advanced Programming Concepts and Practices, Data Structures & Algorithms, Software Architecture, Applied System Design, Machine Learning: Methods of Implementation, Data Mining, Event Driven Programming.

Professional Experience

AMCS Group

Limerick, Co. Limerick

Associate Software Developer

June 2025 – August 2025

- Designed and delivered production Stripe Card and SEPA payment integrations end-to-end, from technical design through implementation, testing, and production deployment.
- Resolved a critical double-charge defect in the live payment flow impacting customer billing accuracy, diagnosing the root cause, patching, and deploying the fix to production.
- Optimised a high-volume transaction data endpoint processing 5,000+ records, reducing response time by 40% from approximately 10 minutes to under 6 minutes.
- Identified and documented the root cause of a serialization bottleneck in the RESTful API layer, producing an analysis that pinpointed the exact failure point and directly enabled the fix.
- Built Azure Data Factory pipelines for ETL processes and managed CI/CD deployments across dev, test, and production Azure environments, handling frequent dev and test releases on a sprint-cadence production deployment cycle.

AMCS Group

Limerick, Co. Limerick

Intern Software Developer

May 2024 – January 2025

- Established unit test coverage across core payment and API components in a large-scale C#/.NET and Angular codebase, writing 500+ tests and consistently maintaining the team's 80% per-PR coverage requirement.
- Shipped bug fixes, feature enhancements, and unit tests to the ACI Cards payment integration throughout an active refactoring cycle, maintaining consistent sprint delivery during concurrent structural changes.
- Traced a data integrity defect in the Payment Express feature to an upstream service misappending results, identifying the root cause of missing column data in the client-facing UI and documenting findings for resolution.
- Extended and maintained 30+ RESTful APIs across live payment and data flows, shipping to production across multiple sprint cycles.

Project Experience

Final Year Project (SimOrch)

University of Limerick

Orchestration Framework for Multi-Agent RE Simulation | [Website](#) | [GitHub](#) | Python | React

September 2025 – April 2026

- Named student researcher on the EU-funded Prompt Me project at Lero, University of Limerick, supervised by Dr. Jacek Dabrowski, investigating the feasibility of AI-driven Requirements Engineering simulation.
- Designed and built a multi-agent orchestration framework in Python coordinating GPT-4-powered Requirements Engineer, Stakeholder, and Analyst Agents with shared memory, persona-based configuration, and structured logging to produce reproducible and traceable RE simulation outputs.
- Evaluated the framework across seven dimensions, achieving 100% feasibility and 100% traceability across all ten simulation

runs, with structural reproducibility of 100% and ground truth requirement coverage of 83.3%.

- Identified a systematic finding across both scenarios: LLM-based RE agents consistently failed to elicit non-functional requirements regardless of persona configuration, pointing to a structural limitation in dialogue-driven elicitation beyond SimOrch specifically.
- Research findings contributed to an in-progress international collaboration spanning University of Limerick, Trinity College Dublin, and The Open University.

Software Development Project (PeakPerformance)

University of Limerick

Sport Marketplace | [Website](#) | [GitHub](#) | Next.js | Supabase | React

February 2025 – April 2025

- Led a team of 4 from design through to deployment, managing task delegation, owning code reviews, and a Kanban-based workflow using branching and pull requests.
- Built a real-time messaging system using Supabase subscriptions and diagnosed and resolved a concurrency bug introduced during feature development that caused intermittent message crashes.
- Developed a full admin dashboard supporting user management (ban/unban), content moderation, message and review removal, and a structured audit log tracking admin actions.
- Built the product listings and discovery experience, implementing server-side search and multi-criteria filtering across marketplace inventory.

Skills

Languages: Java, Python, C#, JavaScript, TypeScript, C++.

Frontend: Next.js, React, Angular, HTML, CSS.

Backend: RESTful APIs, Spring Boot, .NET Framework, Entity Framework, SQL, PostgreSQL, Supabase.

Testing: Postman, NUnit, JUnit, Jasmine, pytest

AI / LLM: LangChain, OpenAI API, Multi-Agent Orchestration.

Infrastructure & Tools: CI/CD, Docker, Azure, Git, Azure Data Factory.

Soft Skills: Problem Solving, Critical Thinking, Communication, Teamwork, Agile Methodologies, Attention to Detail, Adaptability.

References

Available upon request.